

An inventor claims to have built a "flying car" that can hover, where the lifting force is provided by two ducted fans. The car weighs 1,000 kg and has a 200 hp (149.14 kW) engine. The ducted fans are 2.13 m in diameter.

Is hovering flight possible?

[A ducted fan can be considered to have an effective area that is twice that of an unducted rotor.]

$$A_e = 2A = 2.26845\pi \text{ (m}^2\text{)}$$

$$T_h = \frac{1}{2}W = 500g = 4905 \text{ (N)}$$

$$\therefore P_{\text{ideal}} = T v_i = \frac{T^{\frac{3}{2}}}{\sqrt{2\rho A}} = 82212.14 \text{ (W)}$$

$$\therefore P_{\text{total}} = 2P_{\text{ideal}} = 164.42 \text{ (kW)} > 149.14 \text{ (kW)}$$

$\therefore$ hovering flight impossible

A blade has an inboard portion of constant chord c_r from the root to a point $r = r_1$, and then a linear taper from r_1 to the tip at $r = 1$ where the chord is c_t .

Show that the equivalent thrust weighted solidity for this blade is given by:

$$\sigma_e = (\sigma_r - \sigma_t)r_1^3 + \left(\frac{\sigma_t - \sigma_r}{1 - r_1}\right) \left(-\frac{1}{4} + r_1^3 - \frac{3}{4}r_1^4\right) + \sigma_t$$

$$b = \frac{N_b c}{\pi R} = \begin{cases} \frac{N_b c_r}{\pi R} = b_r & \text{for } 0 \leq r \leq r_1 \\ b_t + \frac{b_t - b_r}{1 - r_1} (r - r_1) & \text{for } r_1 \leq r \leq 1 \end{cases}$$

$$C_T = \frac{1}{2} b_e \int_0^1 C_L r^2 dr$$

$$= \frac{1}{2} \left(\int_0^{r_1} b_r C_L r^2 dr + \int_{r_1}^1 \left[b_t + \frac{b_t - b_r}{1 - r_1} (r - r_1) \right] C_L r^2 dr \right)$$

$$\therefore b_e \int_0^1 r^2 dr = b_r \int_0^{r_1} r^2 dr + b_t \int_{r_1}^1 r^2 dr + \frac{b_t - b_r}{1 - r_1} \int_{r_1}^1 r^3 - r^2 dr$$

$$\therefore \frac{1}{3} b_e = \frac{1}{3} b_r r_1^3 + \frac{1}{3} b_t (1 - r_1^3) + \frac{b_t - b_r}{1 - r_1} \left(\frac{1}{4} - \frac{1}{3} - \frac{1}{4} r_1^4 + \frac{1}{3} r_1^3 \right)$$

$$\therefore b_e = (b_r - b_t) r_1^3 + \left(\frac{b_t - b_r}{1 - r_1} \right) \left(-\frac{1}{4} + r_1^3 - \frac{3}{4} r_1^4 \right) + b_t$$